The self-coupled dirac16complex in gravitational fields: symbols and their definitions

A dictionary of every symbol used for the nine-channel self-coupling, the Fierz matrix, the covariant nonlinear field equation, chirality in a gravitational field and the Kohn–Sham theory in the three epochs — sections 15–16 of the notebook 7+1.nb

Repository: github.com/once-ere/7plus1 — prepared for Nash
Notebook: 7+1.nb (Wolfram Language 15.0.1)
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Abstract

For each symbol this dictionary gives the mathematical notation, the name used in the Mathematica notebook 7+1.nb (where there is one), the defining equation and its meaning, grouped by topic in the order of introduction. Longer defining equations follow each table as numbered displayed formulas. Every equation needed to define a symbol is written in this document.

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1. Conventions

- Units: $\hbar = c = 1$; a prime is d/dx_0.
- Coordinates $X = (x_0, x_1, \ldots, x_7)$ (`X = {x0, …, x7}`): x_0 time, x_1, x_2, x_3 ordinary space, $x_4, \ldots, x_7$ auxiliary space.
- Frame indices $A, B \in \{0, \ldots, 7\}$, raised and lowered with $\eta = \mathrm{diag}(+1, -1, \ldots, -1)$ (`η17`), simply the squares of the octonion basis ($\eta_{AB} = \mathrm{Re}(e_A e_B)$); coordinate indices $\mu, \nu \in \{0, \ldots, 7\}$; $i \in \{1, 2, 3\}$ ordinary, $k \in \{4, \ldots, 7\}$ auxiliary space. Repeated indices are summed.
- Multi-indices K, J are strictly increasing lists of frame indices; $|K|$ is the rank (the length of K); $K \in r$ means $|K| = r$.
- Matrices: 1_n identity, M^{T} transpose, $M^{\dagger}$ conjugate transpose, tr trace, $[X, Y] = XY - YX$.
- Mathematica names with Greek letters are written with named characters: `\[CapitalTheta]up` is `Θup`, `\[Eta]K` is `ηK`, `\[Chi]17` is `χ17`, `\[Omega]17` is `ω17`.
- Signature: one time-like dimension (x_0) and seven space-like dimensions; the xAct list $\{7, 1, 0\}$ declares seven space-like dimensions and one time-like dimension, the matrix η lists the squares of the octonion basis (below).

The signature: $\{7,1,0\}$ and $\mathrm{diag}(+1,-1,\ldots,-1)$. The 7+1 spacetime has one time-like dimension (x_0, the direction of the octonion unit $e_0 = 1$) and seven space-like dimensions (x_1, x_2, x_3 ordinary space, $x_4, \ldots, x_7$ auxiliary space). Two different objects describe this, and neither may be replaced by the other. (i) The tensor package xAct, used in section 12 of 7+1.nb exactly as in the author's notebook Einstein-Lovelock-7+1-Nash.nb, declares the metric with `DefMetric[{7, 1, 0}, g71[-a, -b], CD, …]`: the list $\{7,1,0\}$ declares seven space-like dimensions, one time-like dimension and no degenerate dimension — the numbers of entries $+1$ and -1 of $\eta_{71} = \mathrm{diag}(-1,+1,\ldots,+1)$ (`η71`), the same spacetime written in the sign convention of 4+4.nb (space-like = $+1$, time-like = -1). (ii) The matrix $\eta = -\eta_{71} = \mathrm{diag}(+1,-1,\ldots,-1)$ (`η17`) is simply the squares of the octonion basis: $e_0^2 = +1$ for the unit $e_0 = 1$ (time) and $e_i^2 = -1$ for the seven imaginary units $e_1, \ldots, e_7$ (space); more generally it is the real part of the product of two basis units, and it gives the quadratic form of the octonions:

$$\eta_{AB} = \mathrm{Re}(e_A e_B), \qquad e_0^2 = +1, \qquad e_i^2 = -1 \quad (i = 1,\ldots,7),$$
$$\mathrm{Re}(x\,x) = x_0^2 - x_1^2 - \cdots - x_7^2 = \eta_{AB}\, x^A x^B \qquad (x = x^A e_A \in \mathbb{O}).$$

η gives the signs of the Clifford relation $\Gamma_A\Gamma_B + \Gamma_B\Gamma_A = 2\,\eta_{AB}\,1_{16}$ of the real 16×16 Dirac matrices: time squares to $+1_{16}$ and each of the seven space directions to -1_{16}, exactly like the octonion units (in the notation $Cl(p,q)$, with p generators squaring to $+1$ and q to -1, the algebra $Cl(1,7) \cong \mathbb{R}(16)$). It also gives the signs of the metric components: $g_{00} > 0$ along the time-like direction, negative squares along the space-like directions. Every component computed by 7+1.nb follows from the component matrix of the metric; the declared list fixes for xAct the sign of the determinant of the metric, `SignDetOfMetric[g71]` $= -1$, which is the same in both sign conventions, since $\det\eta = \det\eta_{71} = -1$. 7+1.nb checks `SignatureOfMetric[g71]` $= \{7,1,0\}$, `SignDetOfMetric[g71]` $= -1$, $\mathrm{Re}(e_A e_B) = \eta_{AB}$ and $\mathrm{Re}(x\,x) = \eta_{AB}x^A x^B$.

2. Dirac matrices, chirality, and the field

symbol	Mathematica	definition	meaning
Γ_A	`Γ17[A]`	(2.1)	real 16×16 Dirac matrices of 7+1 spacetime
Γ^A	`Γ17up[A]`	$\eta^{AA}\Gamma_A$ (no sum)	
τ_i^O	`τO[i]`	real antisymmetric 8×8, $(\tau_i^O)^2 = -1_8$, anticommuting	the seven imaginary octonion units as matrices
ω	`ω17`	$\Gamma_0\Gamma_1\cdots\Gamma_7$	volume element; $\omega^2 = -1_{16}$
χ	`χ17`	$i\omega$	chirality; $\chi^2 = 1_{16}$, Hermitian
$P_{L,R}$	`PL17`, `PR17`	$\frac{1}{2}(1_{16} \mp \chi)$	chiral projections
E_L, E_R	`EL`, `ER`	(2.2)	chiral basis (16×8)
ϕ_L, ϕ_R	`phiLc`, `phiRc`	$E_{L,R}^\dagger\psi$	8-component chiral parts
ψ	`dirac16complex`; `psN`, `psR`	16 complex anticommuting components	the field; in the checks random Gaussian-integer spinors
$\bar\psi$	`pbS`, `pbR`	$\psi^\dagger\Gamma_0$ (independent symbols in the checks)	Dirac adjoint
α_j, β	`αD[j]`, `βD`	$\Gamma^0\Gamma^j$, Γ_0	Hamiltonian Dirac matrices
m	`mD`; `mks`		mass of dirac16complex ($m = 1$ in section 16)

$$\Gamma_0 = \begin{pmatrix} 0 & 1_8 \\ 1_8 & 0 \end{pmatrix}, \qquad \Gamma_i = \begin{pmatrix} \tau_i^O & 0 \\ 0 & -\tau_i^O \end{pmatrix}, \qquad \Gamma_A\Gamma_B + \Gamma_B\Gamma_A = 2\eta_{AB}1_{16}, \tag{2.1}$$

$$E_L = \frac{1}{\sqrt{2}}\begin{pmatrix} 1_8 \\ -i\,1_8 \end{pmatrix}, \qquad E_R = \frac{1}{\sqrt{2}}\begin{pmatrix} -i\,1_8 \\ 1_8 \end{pmatrix}, \qquad \chi E_L = -E_L, \quad \chi E_R = E_R. \tag{2.2}$$

3. Gravity

symbol	Mathematica	definition	meaning
$e^A{}_\mu$, $e_A{}^\mu$	`fd["e"]`, `fd["ei"]`	$g_{\mu\nu} = \eta_{AB} e^A{}_\mu e^B{}_\nu$, $e_A{}^\mu e^A{}_\nu = \delta^\mu_\nu$	frame and inverse frame
$\Gamma^\mu(X)$	`fdCos["GamUp"]`	$e_A{}^\mu \Gamma^A$	generalized frame matrices
$\{^\lambda_{\mu\nu}\}$	`cv["chr"]`	(3.1)	Christoffel symbols (never written Γ)
$\omega_\mu{}^A{}_B$	`fd["om"]`	(3.2)	canonical spin connection
Ω_μ	`fdCos["Om"]`	(3.2)	spinor connection
D_μ	—	$\partial_\mu + \Omega_\mu$	covariant derivative of spinors
a, b	`a3[x0]`, `b4[x0]`; `aE`, `bE`	(3.3)	scale factors of ordinary and auxiliary space
H_a, H_b	`Ha`, `Hb`; `hE`, `kE`	a'/a, b'/b	Hubble rates
H_a', H_b'	`dHa`, `dHb`	dH_a/dx_0, dH_b/dx_0	
$E^{(k)\mu}{}_\nu$	`ECos[[k+1]]`	(3.4)	Lovelock tensors ($E^{(1)}$ = Einstein tensor)
α_k	`alf[k]`	$\alpha_1 = 1$	Lovelock couplings (α_2, α_3)
$\bar\Lambda$	`lamb`		cosmological constant
κ	`kap`	$8\pi G_8$	gravitational coupling
p, q	`pKs`, `qKs`	$p = (1+2\sqrt{2})/7$, $q = (2-3\sqrt{2})/14$	Einstein–Kasner exponents: $a = x_0^p$, $b = x_0^q$

$$\{^\lambda_{\mu\nu}\} = \tfrac{1}{2} g^{\lambda\rho}\big(\partial_\mu g_{\rho\nu} + \partial_\nu g_{\rho\mu} - \partial_\rho g_{\mu\nu}\big), \tag{3.1}$$

$$\omega_\mu{}^A{}_B = e^A{}_\nu\big(\partial_\mu e_B{}^\nu + \{^\nu_{\mu\lambda}\} e_B{}^\lambda\big), \qquad \Omega_\mu = \tfrac{1}{4}\eta_{AC}\,\omega_\mu{}^C{}_B\,\Gamma^A\Gamma^B, \tag{3.2}$$

$$ds^2 = dx_0^2 - a^2\big(dx_1^2 + dx_2^2 + dx_3^2\big) - b^2\big(dx_4^2 + \cdots + dx_7^2\big), \tag{3.3}$$

$$E^{(k)\mu}{}_\nu = -\frac{1}{2^{k+1}}\,\delta^{\mu\alpha_1\beta_1\cdots\alpha_k\beta_k}_{\nu\gamma_1\delta_1\cdots\gamma_k\delta_k} R^{\gamma_1\delta_1}{}_{\alpha_1\beta_1}\cdots R^{\gamma_k\delta_k}{}_{\alpha_k\beta_k}, \quad \delta^{\mu_1\cdots\mu_n}_{\nu_1\cdots\nu_n} = \det\big[\delta^{\mu_i}_{\nu_j}\big]. \tag{3.4}$$

4. The nine channels and the Fierz matrix

symbol	Mathematica	definition	meaning
K	`mIdx`	the 256 increasing subsets of $\{0,\ldots,7\}$	multi-indices
$r = \lvert K\rvert$	`rk[K]`	length of K	rank, $0 \le r \le 8$
Γ^K	—	$\Gamma^{A_1}\cdots\Gamma^{A_r}$, $\Gamma^{()} = 1_{16}$	product of Dirac matrices
Θ^K	`0up[K]`	(4.1)	Hermitian-phased products; $\bar\psi\Theta^K\psi$ real
η_K	`ηK[K]`	$\eta_{A_1A_1}\cdots\eta_{A_rA_r}$	sign of the multi-index
Θ_K	`0dn[K]`	$\eta_K\Theta^K$	index lowered; $\Theta_K\Theta^K = 1_{16}$
I_r	`Ichan[r, pb, ps]`	(4.2)	Lorentz invariant of rank r
g_r	`gg[r]`, `gSymb`		the nine couplings, $r = 0,\ldots,8$
g_S, g_V, g_A, g_P	`gS`, `gV`, `gA`, `gP`	$g_0 = g_S$, $g_1 = g_V$, $g_7 = -g_A$, $g_8 = -g_P$	the four couplings of section 8 of 7+1.nb
$\mathcal{L}_{\rm self}$	`LselfS`	$-\frac{1}{2}\sum_r g_r I_r$	self-coupling Lagrangian
S	`sLL`	$\exp\big(\frac{1}{8} w_{AB}[\Gamma^A,\Gamma^B]\big)$, $w_{AB} = -w_{BA}$ real	local Lorentz transformation of spinors
F_{rs}	`fierzF`	(4.3)	9×9 Fierz matrix, $F^2 = 1$
R	`RrF`	a 16×16 matrix	density matrix (random in the checks)
D_s	`dS9[R]`	$\sum_{J\in s}\mathrm{tr}(\Theta_J R)\,\mathrm{tr}(\Theta^J R)$	direct (Hartree) invariants

symbol	Mathematica	definition	meaning
G_s	`Geff`	$G = (1 - F^{\mathsf{T}})g$	Hartree–Fock (Fierz-transformed) couplings
G_S, G_V	`gSks`, `gVks`	$G_S = G_0$, $G_V = G_1$	couplings of uniform matter (-100000, 10000)
ρ_S, ρ	`rhoSs`, `rhos`	$\bar\psi\psi$, $\psi^\dagger\psi$ (expectation values)	scalar and number densities

$$\Theta^K = i^{\sigma_r}\,\Gamma^{A_1}\cdots\Gamma^{A_r}, \qquad \sigma_r = \tfrac{r(r-1)}{2} \bmod 2, \tag{4.1}$$

$$I_r = \sum_{|K|=r} (\bar\psi\Theta_K\psi)(\bar\psi\Theta^K\psi), \tag{4.2}$$

$$F_{rs} = \frac{1}{16}\sum_l (-1)^{rs-l}\binom{s}{l}\binom{8-s}{r-l} = \frac{1}{256}\sum_{K\in r}\operatorname{tr}\big(\Theta_K\Theta_J\Theta^K\Theta^J\big) \quad (|J|=s). \tag{4.3}$$

The Fierz identity for commuting spinors a, b, c, d:

$$\sum_{K\in r}(a\Theta_K b)(c\Theta^K d) = \sum_s F_{rs}\sum_{J\in s}(a\Theta_J d)(c\Theta^J b).$$

5. Field equation and Kohn–Sham quantities

symbol	Mathematica	definition	meaning
$\Sigma(X)$	—	(5.1)	Kohn–Sham self-energy (direct minus exchange)
φ_n	—	$(i\Gamma^\mu D_\mu - m - \Sigma)\varphi_n = 0$	Kohn–Sham orbitals
$\mathcal{R}$	—	$\sum_{n\ \mathrm{occ}}\varphi_n\bar\varphi_n$	Kohn–Sham density matrix
$k = (\vec p, \vec q)$	—	comoving momentum	$\vec p$ ordinary, $\vec q$ auxiliary
k_F	`kF`	radius of the comoving Fermi ball	Fermi momentum
$k_{F,\mathrm{sat}}$	`kFsat`	$p = 0$ at $a = b = 1$	saturation Fermi momentum
θ	—	$\lvert k_a\rvert = k\cos\theta$, $\lvert k_b\rvert = k\sin\theta$	angle between ordinary and auxiliary momentum
ω (energy)	`om`	$\sqrt{k^2\cos^2\theta/a^2 + k^2\sin^2\theta/b^2 + M^2}$	single-particle energy (not the volume element)
M, m^*	`Mx`, `ksGap`	$M = m + G_S\rho_S(M)$	effective mass (gap equation)
V_0	—	$G_V\rho$	vector potential
$\langle f\rangle$	`ksMoments`	(5.2)	average over the comoving Fermi ball
ε, p_a, p_b	`ksStress`	(5.3)	energy density, pressures along ordinary / auxiliary space
Π	`pint`	(5.3)	interaction pressure
ε_F	—	$G_V\rho + \sqrt{k_F^2 + M^2}$	Fermi energy (flat spacetime)
E/A	—	ε/ρ	energy per particle
$T^\mu{}_\nu$	—	$\mathrm{diag}(\varepsilon, -p_a, -p_a, -p_a, -p_b, \ldots, -p_b)$	energy–momentum tensor of the Kohn–Sham matter
—	`evolveEL[α2, α3, κ, Ha0]`	returns $\{H_b(0), \{a, b, H_a, H_b\}(20)$, violation$\}$	self-consistent Einstein–Lovelock + Kohn–Sham evolution

$$\Sigma = \sum_r g_r \sum_{|K|=r}\big[\operatorname{tr}(\Theta_K\mathcal{R})\,\Theta^K - \Theta^K\mathcal{R}\,\Theta_K\big], \tag{5.1}$$

$$\langle f\rangle = \frac{1}{2\pi^4 a^3 b^4}\int_0^{k_F} dk\, k^6 \int_0^{\pi/2} d\theta\, \cos^2\theta \sin^3\theta\, f, \qquad \rho = \langle 1\rangle, \quad \rho_S = \langle M/\omega\rangle, \tag{5.2}$$

$$\varepsilon = \langle\omega\rangle - \frac{(M-m)^2}{2G_S} + \frac{G_V}{2}\rho^2, \quad \Pi = \frac{(M-m)^2}{2G_S} + \frac{G_V}{2}\rho^2, \tag{5.3}$$

$$p_a = \left\langle \frac{k^2\cos^2\theta}{3a^2\omega}\right\rangle + \Pi, \quad p_b = \left\langle \frac{k^2\sin^2\theta}{4b^2\omega}\right\rangle + \Pi.$$

6. Symbols that look alike

symbol	**not to be confused with**
$\{7,1,0\}$ (xAct signature list: seven space-like dimensions, one time-like)	$\eta = \mathrm{diag}(+1,-1,\ldots,-1)$ (simply the squares of the octonion basis, $\eta_{AB} = \mathrm{Re}(e_A e_B)$)
$\omega = \Gamma_0\cdots\Gamma_7$ (volume element)	ω (single-particle energy), $\omega_\mu{}^A{}_B$ (spin connection)
χ (chirality)	χ_k (rescaled mode, $\psi = (a^3b^4)^{-1/2}\chi_k$)
Θ^K (upper multi-index)	$\Theta_K = \eta_K\Theta^K$ (lower multi-index)
g_r (bare couplings)	G_s (Fierz-transformed couplings), $g_{\mu\nu}$ (metric)
F_{rs} (Fierz matrix)	$\mathcal{F}_{\mu\nu}$ (spin curvature, not used here)
R (density matrix)	$\mathcal{R}$ (Kohn–Sham density matrix), $R^{\gamma\delta}{}_{\alpha\beta}$ (Riemann tensor)
p (Kasner exponent; pressure in flat spacetime)	p_a, p_b (pressures), $\vec{p}$ (ordinary momentum)
q (Kasner exponent)	$\vec{q}$ (auxiliary momentum)
α_k (Lovelock couplings)	α_j (Dirac matrices $\Gamma^0\Gamma^j$)
$\kappa = 8\pi G_8$	—